THE OHIO STATE UNIVERSITY
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Integrating Robotics & Engineering Education: Initial Validation of an *SCCT*- and *CT*-Based Survey Instrument

Work-in-Progress · Evidence-Based Practice · Initial validity evidence for a robotics-based engineering self-efficacy scale

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01 INTRODUCTION

Robotics education fosters positive STEM attitudes and strengthens students' computational thinking, while pre-college engineering education increases self-efficacy and interest in engineering. Combining the two could more effectively prepare students for future engineering study and careers.

Yet research examining their integration—using robotics projects as a vehicle for engineering learning—remains limited, and most existing implementations are short in duration with few rigorous evaluations.

This gap calls for validated instruments to measure the impact of robotics-integrated engineering courses.

02 THE E4USA+FIRST INITIATIVE

A pre-college engineering initiative expanding nationwide access to robotics-based engineering learning. Flexible lessons serve as a stand-alone curriculum or supplemental content in high-school STEM courses.

Through partnerships between Engineering for US All (e4usa) and FIRST, the initiative positions teachers as co-creators and engages diverse urban, suburban, and rural contexts to strengthen STEM literacy and broaden the future engineering workforce.

03 THEORETICAL FRAMEWORK

SCCT — SOCIAL COGNITIVE CAREER THEORY

Explains how self-efficacy, outcome expectations, and goals jointly shape individuals' academic and career interests and choices [1].

CT — COMPUTATIONAL THINKING

A problem-solving framework involving abstraction, decomposition, and algorithmic reasoning to formulate solutions [2].

04 PURPOSE

This work-in-progress study provides initial validity evidence for an adapted survey instrument measuring robotics-based engineering self-efficacy in secondary students—testing whether the scale is a coherent, reliable measure for the e4usa+FIRST context.

05 INSTRUMENT DESIGN

The full adapted survey integrates SCCT and CT to examine the impact of the e4usa+FIRST curriculum on high-school students' engineering competencies. All items use a 4-point Likert scale, from **0 = no confidence** to **3 = complete confidence**.

SCCT COMPONENT — 5 CONSTRUCTS · 30 ITEMS

Confidence in robotics-based engineering design · outcome expectations · engineering/robotics interest · intention & commitment to future plans · perceived support & barriers.

CT COMPONENT — 5 CONSTRUCTS · 20 ITEMS

Creativity · algorithmic thinking · cooperativity · critical thinking · problem solving.

THIS STUDY → SELF-EFFICACY CONSTRUCT · 8 ITEMS

Confidence in robotics-based engineering design, adapted from a previously validated pre-college engineering self-efficacy scale [4].

THE 8-ITEM SELF-EFFICACY SCALE

"I am confident I can..." (across the engineering design process)

- Identify a design need for a robotics-based solution
- Research that design need
- Develop robotics-based engineering solutions
- Select the best possible design
- Construct a prototype
- Evaluate and test a design
- Communicate a design
- Redesign a solution

06 METHODS

Pre-survey data from **81 valid responses** across **7 schools** (2025–26 academic year; OSU IRB-approved). Validity and reliability analyzed in four stages, using Python (*pandas*, *factor_analyzer*):

- Descriptive statistics & distribution screening
- Factorability — Kaiser-Meyer-Olkin (KMO) + Bartlett's Test of Sphericity
- Exploratory Factor Analysis — Principal Axis Factoring (Kaiser criterion + scree plot) [5]
- Internal consistency — Cronbach's alpha (α)

SAMPLE CHARACTERISTICS

81 high-school students from seven participating schools. Gender: **54 male (66.7%)**, **23 female (28.4%)**, 4 not reported. The sample was racially and ethnically diverse:

RACE / ETHNICITY	N	%
White	25	30.9
Multiracial	19	23.5
Asian	14	17.3
Middle Eastern	12	14.8
Black / African American	5	6.2
Hispanic / Latino	4	4.9
Not reported	2	2.5
Total	*81*	*100*

BY THE NUMBERS

81 STUDENTS

7 SCHOOLS

50 ITEMS · 10 CONSTRUCTS

8 ITEMS VALIDATED HERE

DESCRIPTIVE STATISTICS · TABLE 2

ITEM (CONFIDENCE TO...)	MEAN	SD
Identify a design need	1.77	0.73
Research a design need	1.93	0.70
Develop solutions	1.69	0.80
Select the best design	1.65	0.82
Construct a prototype	1.75	0.86
Evaluate and test a design	2.05	0.77
Communicate a design	1.73	0.81
Redesign a solution	*1.59*	*0.82*

Means **1.59–2.05** (0–3 scale) — moderate confidence; slight left-skew, minimal missing data.

07 RESULTS: A VALID, RELIABLE SCALE

0.845 KMO · MERITORIOUS

0.85 CRONBACH'S A

Factorability was strong (KMO = 0.845 > 0.80; Bartlett's test $p < 0.001$). EFA supported a single-factor structure, with all eight loadings ranging **0.47–0.80** (> 0.40) — evidence of convergent validity. Internal consistency was high (α = 0.85).

FACTOR LOADINGS (SINGLE FACTOR)

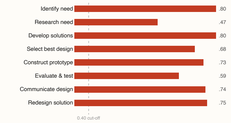


FIG 1. EFA factor loadings (Principal Axis Factoring) for the 8 self-efficacy items.

08 CONCLUSION & FUTURE WORK

The adapted robotics-based engineering self-efficacy scale shows initial evidence of reliability and conceptual coherence for secondary students. Future work will test relationships among SCCT and CT constructs and the stability of the factor structure in larger, more diverse samples—supporting impact-focused research on robotics-integrated engineering education.

SELECTED REFERENCES

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